Continuity and the Differential Quotient

By *Ernst Steinitz* in Charlottenburg.

That there are continuous functions which possess a differential quotient at no point was first proved by *Weierstrass*. Other authors have added a series of further examples to those given by him. In what follows, some general considerations on continuity and differentiability of functions will be set out, from which the existence of continuous non-differentiable functions arises quite naturally, so that the ordinary character of the case is brought out in a clear way and, I hope, in a form suitable for teaching. Throughout, only finite real single-valued functions of one real variable will be considered.

§ 1.

Let p and q be two real numbers, $q > p$; let A be the domain of all numbers between p and q inclusive; and let $f(x)$ be a continuous real single-valued function defined on this domain. Then $f(x)$ is also uniformly continuous. If

$$a_0, a_1, a_2, \ldots$$

is a fundamental sequence of numbers in A, then its limit a also belongs to the domain A, and from the continuity of $f(x)$ it follows that the corresponding function values

$$f(a_0), f(a_1), f(a_2), \ldots$$

form a fundamental sequence with limit $f(a)$.

From this we can draw an important conclusion. Let B be a numerical domain contained in A (so that every number of B is also in A), and suppose it has the property that every number of A can be represented as the limit of a fundamental sequence of numbers from B. It follows that the continuous function $f(x)$ is determined on the whole domain A by the values which it assumes at the points of the domain B. Thus, if $\varphi(x)$ is a function defined on the domain B, there can certainly be no more than one function $f(x)$ whose domain of validity is A, which everywhere there satisfies the condition of continuity, and which agrees with $\varphi(x)$ at the points of B.

Of course, not every function $\varphi(x)$ defined on the domain B will admit a function $f(x)$ of the indicated kind. For if such a function exists, then it follows that $\varphi(x)$ has the following property: if $\varepsilon > 0$ is an arbitrarily small quantity, one can choose $\delta > 0$ so that for any two points x_1 and x_2 of the domain B whose difference is $< \delta$, the difference of the corresponding function values $\varphi(x_2)$ and $\varphi(x_1)$ is $< \varepsilon$. Conversely, if $\varphi(x)$ satisfies this condition, it is also easy to prove that a function $f(x)$ of the indicated kind exists. Indeed, if a is any point of A, then the assumption made about $\varphi(x)$ shows that to every fundamental sequence of points of B

$$b_0, b_1, b_2, \ldots$$

with limit a there corresponds a fundamental sequence

$$\varphi(b_0), \varphi(b_1), \varphi(b_2), \ldots,$$

and that all these fundamental sequences have the same limit. Denoting this limit by $f(a)$, one obtains a uniquely defined function $f(x)$ on the entire domain A, and it is easily seen that it is continuous and agrees with $\varphi(x)$ at the points of B.

I shall say that a function φ, defined on any domain, is continuous within this domain if, for any two points x_1, x_2 of the domain, as soon as their difference is sufficiently small, the difference

$\varphi(x_2) - \varphi(x_1)$ remains below a prescribed magnitude. It is immaterial here whether the domain of validity of φ itself is continuous or not. The result just obtained can then be stated as follows:

The necessary and sufficient condition for the existence of a function $f(x)$, continuous in the domain A, which agrees at all points of the domain B with a function $\varphi(x)$ defined there, is the continuity of the function $\varphi(x)$ within B.

§ 2.

To have a more definite picture before us, choose as the domain A the numbers between 0 and 1, including the endpoints. Let m be a positive integer > 1, and let B be the totality of the rational fractions in A whose denominator is a power of m. Since every number between 0 and 1 can be represented as the limit of a fundamental sequence formed from such fractions, the considerations made in § 1 apply; and in order to obtain all functions $f(x)$ continuous in A, it is only necessary to construct the functions $\varphi(x)$ continuous in B, to each of which a definite function $f(x)$ corresponds. In contrast to the numbers of A, the numbers of B constitute a countable set, and it is on this fact that the introduction of the domain B in place of A actually makes it easier to survey the totality of continuous functions.

We shall restrict ourselves to functions which vanish at $x = 0$; all others may be derived from these by adding a constant. Let $\varphi(x)$ be a function of this kind, defined on the domain B, which at first need not be continuous. The numbers of B may be arranged by first dividing the interval $0 \dots 1$ into m, then into m^2, m^3 equal parts, and so on, and writing down the division-points obtained in their natural order, as the following scheme shows:

$$\begin{array}{ccccccccc}
0 & & & & 1 & & & & \\
0 & & \frac{1}{m} & & \frac{2}{m} & \cdots & 1 & & \\
0 & \frac{1}{m^2} & \frac{2}{m^2} & \cdots & \frac{1}{m} & \frac{m+1}{m^2} & \cdots & \frac{2}{m} & \cdots 1 \\
\cdots & & & & & & & &
\end{array}$$

Put

$$\varphi(1) - \varphi(0) = \Delta_{0,0},$$

$$\varphi\left(\frac{1}{m}\right) - \varphi(0) = \Delta_{1,0}, \quad \varphi\left(\frac{2}{m}\right) - \varphi\left(\frac{1}{m}\right) = \Delta_{1,1}, \dots, \varphi(1) - \varphi\left(\frac{m-1}{m}\right) = \Delta_{1,m-1},$$

and in general

$$\varphi\left(\frac{l+1}{m^k}\right) - \varphi\left(\frac{l}{m^k}\right) = \Delta_{k,l} \qquad \begin{array}{c} k = 0, 1, \dots, \infty, \\ l = 0, 1, \dots, m^k - 1. \end{array} \tag{1}$$

Then not only are all the differences $\Delta_{k,l}$ determined by the function $\varphi(x)$, but conversely, since $\varphi(0) = 0$ has been assumed, the function $\varphi(x)$ is completely determined by the differences $\Delta_{k,l}$. From (1) it follows that

$$\Delta_{k,l} = \Delta_{k+1,lm} + \Delta_{k+1,lm+1} + \cdots + \Delta_{k+1,lm+m-1}. \tag{2}$$

Thus, if one wants to construct a function $\varphi(x)$ valid on the domain B by prescribing its value differences $\Delta_{k,l}$, one must proceed as follows. Choose $\Delta_{0,0}$ arbitrarily; split $\Delta_{0,0}$ in any manner into m parts $\Delta_{1,0}, \Delta_{1,1}, \dots, \Delta_{1,m-1}$ so that $\Delta_{1,0} + \cdots + \Delta_{1,m-1} = \Delta_{0,0}$; then split each difference Δ so obtained again into m parts, and continue in infinitum, as prescribed by (2). Then the function φ is defined on B by the condition $\varphi(0) = 0$ and the equations (1), compatible with one another by (2). Naturally, if a function $\varphi(x)$ is to be constructed by the indicated method, some rule must be given according to which the subdivision of the quantities Δ is to be carried out.

The function $\varphi(x)$ thus obtained will in general not be continuous. But if it is continuous, then by means of it we pass to a function $f(x)$ valid on the whole domain A and continuous there, as was shown in § 1.

§ 3.

In order to lead to a function φ valid on the domain B, the differences Δ must be chosen according to the conditions of § 2, (2). The question now arises: what further conditions must be added if the function φ is to turn out continuous?

Let α_k $(k = 0, 1, \ldots)$ denote the greatest of the absolute values of

$$\Delta_{k,0}, \Delta_{k,1}, \ldots, \Delta_{k,m^k-1}.$$

Then α_k is an increment of the function φ corresponding to an increment of the variable by $1/m^k$; and since $\lim_{k=\infty} 1/m^k = 0$, it follows as a necessary condition for the continuity of $\varphi(x)$ that one must also have

$$\lim_{k=\infty} \alpha_k = 0. \tag{1}$$

But that this condition is not sufficient is easily shown by examples. It is not at all difficult to translate the condition of continuity of φ into conditions on the differences Δ, but these conditions do not take a perspicuous form. I shall therefore restrict myself to giving a very general condition under which one is certainly led to a continuous function. I assert:

If the sequence of quantities just defined,

$$\alpha_0, \alpha_1, \alpha_2, \ldots,$$

which we may call the maximal differences, satisfies not only condition (1) but the stronger condition that their sum

$$\sum_{k=0}^{\infty} \alpha_k \tag{2}$$

converges, then the function φ defined by the equations § 2, (1) is continuous.

To prove this assertion, assume the convergence of the sum (2) and set

$$\sum_{k=i}^{\infty} \alpha_k = \mu_i \qquad (i = 0, 1, \ldots). \tag{3}$$

Then

$$\lim_{i=\infty} \mu_i = 0. \tag{4}$$

If p/m^k and q/m^k are any two numbers in B, then from the definition of the quantities α one obtains

$$\left|\varphi\Big(\frac{q}{m^k}\Big) - \varphi\Big(\frac{p}{m^k}\Big)\right| \leq |q - p|\alpha_k, \tag{5}$$

where the vertical bars mean that the absolute value of the enclosed quantity is to be taken. If x is a number of B lying between l/m^k and $(l + 1)/m^k$,

$$\frac{l}{m^k} \leq x \leq \frac{l+1}{m^k}, \tag{6}$$

then x has the form

$$x = \frac{s}{m^{k+r}} \qquad (r \geq 0), \tag{7}$$

and, if $x < \frac{l+1}{m^k}$, one may write

$$x = \frac{l}{m^k} + \frac{c_1}{m^{k+1}} + \frac{c_2}{m^{k+2}} + \cdots + \frac{c_r}{m^{k+r}}, \tag{8}$$

and if $\frac{l}{m^k} < x$, one may write

$$x = \frac{l+1}{m^k} - \left(\frac{c'_1}{m^{k+1}} + \frac{c'_2}{m^{k+2}} + \cdots + \frac{c'_r}{m^{k+r}}\right), \tag{9}$$

where the coefficients c, c' are integers satisfying

$$0 \leq c \leq m-1, \qquad 0 \leq c' \leq m-1. \tag{10}$$

For abbreviation set

$$\frac{l}{m^k} = x_0, \quad \frac{l+1}{m^k} = x'_0,$$

$$\frac{l}{m^k} + \frac{c_1}{m^{k+1}} + \cdots + \frac{c_h}{m^{k+h}} = x_h, \quad \frac{l+1}{m^k} - \left(\frac{c'_1}{m^{k+1}} + \cdots + \frac{c'_h}{m^{k+h}}\right) = x'_h,$$

$$(h = 1, \ldots, r),$$

so that $x_r = x = x'_r$. From the corresponding expressions for $x_{h-1}, x_h, x'_{h-1}, x'_h$, and taking account of (5) and (10), it follows that

$$|\varphi(x_h) - \varphi(x_{h-1})| \leq c_h \alpha_{k+h} \leq (m-1)\alpha_{k+h},$$

$$\left|\varphi(x'_h) - \varphi(x'_{h-1})\right| \leq c'_h \alpha_{k+h} \leq (m-1)\alpha_{k+h},$$

and hence by (3)

$$\left|\varphi(x) - \varphi\left(\frac{l}{m^k}\right)\right| = |\varphi(x_r) - \varphi(x_0)| \leq (m-1)\mu_{k+1} \leq (m-1)\mu_k,$$

$$\left|\varphi(x) - \varphi\left(\frac{l+1}{m^k}\right)\right| = |\varphi(x'_r) - \varphi(x'_0)| \leq (m-1)\mu_{k+1} \leq (m-1)\mu_k.$$

The inequalities

$$\left|\varphi(x) - \varphi\left(\frac{l}{m^k}\right)\right| \leq (m-1)\mu_k, \quad \left|\varphi(x) - \varphi\left(\frac{l+1}{m^k}\right)\right| \leq (m-1)\mu_k \tag{11}$$

also hold when $x = l/m^k$ or $x = (l+1)/m^k$.

Now let x_1 and $x_2 \geq x_1$ be two numbers in B whose difference is $< 1/m^k$. Choose the integer l so that

$$\frac{l}{m^k} \leq x_1 \leq \frac{l+1}{m^k} \tag{12}$$

and thus by (11)

$$\left|\varphi(x_1) - \varphi\left(\frac{l+1}{m^k}\right)\right| \leq (m-1)\mu_k. \tag{13}$$

Then x_2 lies either between l/m^k and $(l+1)/m^k$ or between $(l+1)/m^k$ and $(l+2)/m^k$. In both cases

$$\left|\varphi(x_2) - \varphi\left(\frac{l+1}{m^k}\right)\right| \leq (m-1)\mu_k, \tag{14}$$

and therefore

$$|\varphi(x_2) - \varphi(x_1)| \leq 2(m-1)\mu_k. \tag{14'}$$

But by (4), $\lim_{k=\infty} 2(m-1)\mu_k = 0$, and thus the continuity of $\varphi(x)$ is proved.

The convergence of the sum (2) gives a sufficient but not necessary condition for the continuity of $\varphi(x)$. Hence, if we construct in all possible ways systems of quantities $\Delta_{k,l}$ satisfying the relations § 2, (2), and for which the sum $\sum_{k=0}^{\infty} \alpha_k$ converges, then each such system does indeed yield a continuous function, but the totality of functions obtained in this way represents only a part of all continuous functions.

§ 4.

The indicated method for producing continuous functions may be illustrated by a simple example.

Choose for $\Delta_{0,0}$ any nonzero number, and split $\Delta_{0,0}$ in any manner into m parts $\Delta_{1,0},\dots,\Delta_{1,m-1}$; assume only that these are all also nonzero. For further subdivision we shall give the following rule: the m parts into which any Δ is split shall stand in the same ratio to one another as the m parts $\Delta_{1,0},\dots,\Delta_{1,m-1}$ into which $\Delta_{0,0}$ was split. This rule determines the entire system of differences Δ. If $\Delta_{k,l}$ is given, then from

$$\Delta_{1,0}+\cdots+\Delta_{1,m-1}=\Delta_{0,0},$$

$$\Delta_{k+1,lm}+\cdots+\Delta_{k+1,lm+m-1}=\Delta_{k,l},$$

$$\Delta_{k+1,lm}:\Delta_{k+1,lm+1}:\cdots:\Delta_{k+1,lm+m-1}=\Delta_{1,0}:\Delta_{1,1}:\cdots:\Delta_{1,m-1},$$

it follows that

$$\Delta_{k+1,lm}=\Delta_{k,l}\frac{\Delta_{1,0}}{\Delta_{0,0}},\quad \Delta_{k+1,lm+1}=\Delta_{k,l}\frac{\Delta_{1,1}}{\Delta_{0,0}},\quad \dots$$

We shall call the indicated method of dividing the Δ a periodic subdivision procedure. For abbreviation set

$$\Delta_{1,0}=\delta_1,\quad \Delta_{1,1}=\delta_2,\quad \dots\quad \Delta_{1,m-1}=\delta_m.$$

Then, under a periodic subdivision procedure, $\delta_1,\dots,\delta_m$ determine a function $\varphi(x)$ valid on B, satisfying the condition $\varphi(0)=0$ and the equations § 2, (1). We denote this function by

$$\varphi(x;\delta_1,\delta_2,\dots,\delta_m).$$

If it is continuous, it corresponds to a continuous function valid on all of A, for which we introduce the notation

$$f(x;\delta_1,\delta_2,\dots,\delta_m).$$

To see when the function $\varphi(x;\delta_1,\dots,\delta_m)$ becomes continuous, one must consider the maximal differences $\alpha_0,\alpha_1,\dots$. In the present case it is immediate that these form a geometric sequence. Thus the condition $\lim\alpha_k=0$ and the condition of convergence of the sum $\alpha_0+\alpha_1+\cdots$ coincide here; they are fulfilled when $\alpha_1<\alpha_0$. We can state the result obtained in this way as follows:

In order that a periodic subdivision procedure lead to a continuous function, it is necessary and sufficient that the m parts into which $\Delta_{0,0}$ is split are all, in absolute value, smaller than $\Delta_{0,0}$ itself.

For $m=2,3,4$ the functions

$$f(x;1,2),\quad f(x;3,3,-2),\quad f(x;1,1,1,-1)$$

are continuous functions formed by a periodic subdivision procedure; for in all three cases one has

$$|\delta_1+\delta_2+\cdots+\delta_m|>|\delta_i|\qquad(i=1,2,\dots,m).$$

Writing $f_1(x),\varphi_1(x)$ for $f(x;1,2),\varphi(x;1,2)$, one obtains

$$f_1(0)=\varphi_1(0)=0,\quad f_1\left(\frac{1}{2}\right)=\varphi_1\left(\frac{1}{2}\right)=1,\quad f_1(1)=\varphi_1(1)=3,$$

$$f_1\left(\frac{1}{4}\right)=\varphi_1\left(\frac{1}{4}\right)=\frac{1}{3},\quad f_1\left(\frac{3}{4}\right)=\varphi_1\left(\frac{3}{4}\right)=\frac{5}{3}.$$

For $f_1(1/3)$, since

$$\frac{1}{3}=\frac{1}{4}+\left(\frac{1}{4}\right)^2+\left(\frac{1}{4}\right)^3+\cdots,$$

one obtains the value

$$\frac{1}{3} + \frac{1}{3}\left(\frac{2}{9}\right) + \frac{1}{3}\left(\frac{2}{9}\right)^2 + \cdots, \qquad f_1\left(\frac{1}{3}\right) = \frac{3}{7}.$$

The computation of the function $f(x; \delta_1, \ldots, \delta_m)$ for definite values of x is therefore very simple. If the numbers $\delta_1, \ldots, \delta_m$ are rational, it is easy to see that for rational x the function always assumes a rational value.

§ 5.

Let $f(x)$ be, as at the beginning, a continuous function defined on the domain A. To every interval $x_1 \ldots x_2$ which forms a part of the interval $0 \ldots 1$ there corresponds a definite difference quotient

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

To determine whether the function $f(x)$ has a differential quotient at the point $x = x_0$, one forms the difference quotient for an interval $x_1 \ldots x_2$ containing the point x_0, and contracts the interval $x_1 \ldots x_2$ to the point x_0. If, for every way in which this contraction can be performed, the difference quotient approaches one and the same finite limit, then this limit is the differential quotient of $f(x)$ at the point $x = x_0$. Otherwise no finite differential quotient exists at this point. If, for example, the interval $x_1 \ldots x_2$ can be contracted in such a way that the absolute value of $\frac{f(x_2)-f(x_1)}{x_2-x_1}$ grows beyond all bounds, then no finite differential quotient is present at the point under consideration. The differential quotient may still have a definite infinitely large value $+\infty$ or $-\infty$, that is, under every manner of contraction of the interval the difference quotient may grow beyond all positive numbers or fall below all negative numbers; but we shall not enter into this at first. We shall diminish the interval enclosing the point x_0 in the following way. For the variable k, which is to run through all positive integers, determine the integer l_k so that x_0 lies between l_k/m^k and $(l_k+1)/m^k$, including the lower endpoint and, if $x_0 \neq 1$, excluding the upper endpoint. If $\varphi(x)$ again denotes the function defined on B and agreeing there with $f(x)$, and if the $\Delta_{k,l}$ have their former meaning, then for

$$x_1 = \frac{l_k}{m^k}, \qquad x_2 = \frac{l_k + 1}{m^k}$$

one has

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = m^k \Delta_{k,l_k}.$$

If

$$\lim_{k=\infty} m^k |\Delta_{k,l_k}| = \infty,$$

then there is no finite differential quotient at the point x_0. Let β_k denote the smallest of the absolute values $|\Delta_{k,0}|, |\Delta_{k,1}|, \ldots, |\Delta_{k,m^k-1}|$. Then $m^k\beta_k$ gives the smallest of the corresponding difference quotients. Hence, if

$$\lim_{k=\infty} m^k \beta_k = \infty,$$

then $f(x)$ has no finite differential quotient at any point.

§ 6.

From the last result one obtains a simple procedure for constructing continuous functions without finite differential quotients. We produce a system of differences $\Delta_{k,l}$ by specifying a procedure according to which the subdivision of the Δ is to be carried out (§ 2). In order to arrive certainly at a continuous function, choose the procedure so that the resulting absolute maximal differences

$$\alpha_0, \alpha_1, \alpha_2, \ldots \tag{1}$$

yield a convergent sum (§ 3). Then the sequence of minimal differences considered in § 5,

$$\beta_0, \beta_1, \beta_2, \ldots, \tag{2}$$

will also have a convergent sum. The same is true in any case of the sequence

$$1, \frac{1}{m}, \frac{1}{m^2}, \ldots. \tag{3}$$

If it is now possible to make the convergence of the second sum so slow in comparison with the convergence of the third that the quotients formed from corresponding terms of (2) and (3),

$$\beta_0, m\beta_1, m^2\beta_2, \ldots, \tag{4}$$

which form the sequence of minimal difference quotients, grow beyond all bounds, then the continuous function obtained in this way is nowhere differentiable (§ 5).

That this can be achieved in many ways is very easy to see. The simplest examples are obtained by using a periodic subdivision procedure (§ 4). Then the sequence (4), like the others, is geometric. The condition $\lim_{k=\infty} m^k\beta_k = \infty$ is fulfilled when $m\beta_1 > \beta_0$, that is, when β_1, the smallest of the differences $|\Delta_{1,0}| \ldots |\Delta_{1,m-1}|$, is greater than $\frac{1}{m}\beta_0 = \frac{1}{m}|\Delta_{0,0}|$. Thus, combining this result with that obtained in § 4, one obtains the following rule: construct, by a periodic subdivision procedure, a system of differences Δ such that the m parts into which $\Delta_{0,0}$ is split are, in absolute value, smaller than $\Delta_{0,0}$ but greater than $\frac{1}{m}\Delta_{0,0}$. Then this system leads to a continuous function which is nowhere differentiable.

Such a subdivision is always possible if $m > 2$. The functions

$$f(x; 3, -2, 3), \quad f(x; 1, 1, 1, -1), \quad f(x; 4, 4, -5, -5, 4, 4)$$

are examples of this.

§ 7.

So far the question of the possible presence of a definite infinitely large differential quotient has been left undiscussed. The method given in § 6 does not exclude the possibility that an infinitely large differential quotient occurs at certain points. We shall now avoid this. If, as is done here, one restricts oneself to functions produced by a periodic subdivision procedure, then one need add to the rules given at the end of § 6 only one more rule (executable for $m > 5$) in order also to make the occurrence of infinitely large differential quotients impossible. This rule is realized by the last function given above, $f(x; 4, 4, -5, -5, 4, 4)$, and may be explained using this example. Here we have

$$m = 6, \qquad \Delta_{0,0} = 6,$$

$$\Delta_{1,0} = 4, \quad \Delta_{1,1} = 4, \quad \Delta_{1,2} = -5, \quad \Delta_{1,3} = -5, \quad \Delta_{1,4} = 4, \quad \Delta_{1,5} = 4.$$

Alongside the differences $\Delta_{1,l}$, consider the sums formed from adjacent such differences, namely the sums

$$\Delta_{1,0}+\Delta_{1,1},\ \Delta_{1,1}+\Delta_{1,2}, \ldots, \Delta_{1,m-2}+\Delta_{1,m-1},\ \Delta_{1,0}+\Delta_{1,1}+\Delta_{1,2}, \ldots, \Delta_{1,0}+\Delta_{1,1}+\Delta_{1,2}+\cdots+\Delta_{1,m-1}.$$

To each difference $\Delta_{1,l}$ we assign one of these sums, denoted by $\Delta'_{1,l}$, in such a way that $\Delta_{1,l}$ itself occurs among the summands of $\Delta'_{1,l}$. In the present example one can choose $\Delta'_{1,l}$ $(l = 0, 1, \ldots, m-1)$ so that $\Delta'_{1,l}$ and $\Delta_{1,l}$ have opposite signs, for instance by setting

$$\begin{aligned} \Delta'_{1,0} &= \Delta_{1,0} + \Delta_{1,1} + \Delta_{1,2} + \Delta_{1,3}, & \Delta'_{1,1} &= \Delta_{1,1} + \Delta_{1,2}, \\ \Delta'_{1,2} &= \Delta_{1,0} + \Delta_{1,1} + \Delta_{1,2}, & \Delta'_{1,3} &= \Delta_{1,3} + \Delta_{1,4} + \Delta_{1,5}, \\ \Delta'_{1,4} &= \Delta_{1,3} + \Delta_{1,4}, & \Delta'_{1,5} &= \Delta_{1,2} + \Delta_{1,3} + \Delta_{1,4} + \Delta_{1,5}. \end{aligned} \tag{1}$$

This gives

$$\Delta'_{1,0} = -2, \quad \Delta'_{1,1} = -1, \quad \Delta'_{1,2} = 3, \quad \Delta'_{1,3} = 3, \quad \Delta'_{1,4} = -1, \quad \Delta'_{1,5} = -2.$$

Whenever such a case occurs, and the conditions stated in the preceding paragraph are also fulfilled, we are dealing, as will now be shown, with a function which possesses neither a finite nor an infinitely large differential quotient at any point.

The quantity $\Delta'_{1,l}$ represents a function increment corresponding to an increment of the variable by ε_l/m, where ε_l denotes the number of summands of which $\Delta'_{1,l}$ is composed. The corresponding difference quotient is

$$\frac{1}{\varepsilon_l} m \Delta'_{1,l} = \mathfrak{D}_l \, m \Delta_{1,l},$$

where $\mathfrak{D}_l$ stands for $\frac{1}{\varepsilon_l} \cdot \frac{\Delta'_{1,l}}{\Delta_{1,l}}$. By the preceding construction, $\mathfrak{D}_0, \mathfrak{D}_1, \ldots, \mathfrak{D}_{m-1}$ are negative numbers. The difference $\Delta_{k,l}$ splits as the sum

$$\Delta_{k,l} = \Delta_{k+1,lm} + \cdots + \Delta_{k+1,lm+m-1}.$$

By

$$\Delta'_{k+1,lm}, \Delta'_{k+1,lm+1}, \ldots, \Delta'_{k+1,lm+m-1}$$

we mean the quantities obtained by replacing, in equations (1), $\Delta_{1,s}, \Delta'_{1,s}$ $(s = 0, \ldots, m-1)$ by $\Delta_{k+1,lm+s}, \Delta'_{k+1,lm+s}$. Then

$$\frac{\Delta'_{k+1,lm+s}}{\Delta_{k+1,lm+s}} = \frac{\Delta'_{1,s}}{\Delta_{1,s}};$$

each $\Delta_{k,l}$ and each $\Delta'_{k,l}$ represents a function increment, and the corresponding difference quotients are

$$m^k \Delta_{k,l}, \qquad \mathfrak{D}_l \, m^k \Delta_{k,l}.$$

Here $\mathfrak{D}_l$ is equal to that one of the numbers $\mathfrak{D}_0, \mathfrak{D}_1, \ldots, \mathfrak{D}_{m-1}$ whose index is congruent to l modulo m.

Let x_0 be an arbitrary point of the domain A, and choose the integer l_k so that $l_k/m^k \le x_0 < (l_k+1)/m^k$. Then not only Δ_{k,l_k} but also Δ'_{k,l_k} corresponds to an interval which encloses the point x_0. The associated difference quotients are

$$m^k \Delta_{k,l_k}, \qquad \mathfrak{D}_{l_k} m^k \Delta_{k,l_k}.$$

By § 6 the first, and hence also the second, grows beyond all bounds in absolute value as k increases. The two quotients also always have opposite signs. Hence at the point x_0 there is neither a finite nor a definite infinitely large differential quotient. Rather, inside every arbitrarily small interval $x'_1 \ldots x'_2$ containing x_0 in its interior, one can determine a second interval $x_1 \ldots x_2$ of the same kind for which the difference quotient

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

assumes any prescribed value. The latter follows from the preceding if one observes that the difference quotient $\frac{f(x_2)-f(x_1)}{x_2-x_1}$, apart from points where $x_1 = x_2$, is a continuous function of the variables x_1, x_2.